

Edge-Deployable EEG Foundation Models: Scaling Laws and Cross-Dataset Transfer for Compact SIGReg-Pretrained Encoders

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Abstract

Self-supervised EEG representation learning has converged on foundation-scale recipes [1, 2, 3, 4] (tens of thousands of hours, 5–10 M+ parameters, multi-GPU). The compact, single-corpus regime in which most laboratories and on-device applications operate is uncharacterized, and standard cross-subject probes can be silently inflated by subject leakage. We address both gaps using **EEG-LeJEPA**, a 2.86 M-parameter Joint-Embedding Predictive Architecture trained with Sketched Isotropic Gaussian Regularization (SIGReg) [5] as a leakage-clean testbed. Pretrained for 10,000 steps on 50 EEGMMIDB subjects (51–100, disjoint from evaluation) in under 15 minutes on a single consumer GPU, it reaches 68.6%/0.751 AUC on rest-versus-activity and 62.1%/0.690 AUC on left-versus-right motor imagery via a frozen-encoder linear probe (LOSO; subjects 1–20 strictly held out); both gains over a random-initialized identical architecture are large and significant (+11.7 and +10.9 pp; Benjamini–Hochberg $q < 0.002$). Three transferable findings emerge. First, under the disjoint split, predictor-derived feature sources lose their apparent advantage and the simple encoder mean-pool is the strongest and most trustworthy source. Second, the SIGReg projection count must scale with embedding width: an isolation run shows a $2.4\times$ larger model collapses 8.3 pp on left-vs-right MI when only M is reduced; at properly-scaled M it is modestly but consistently better across three seeds (+1.3 to +2.0 pp). Third, channel-padded cross-dataset transfer outperforms channel-matched re-pretraining (+4.9 vs. +1.5 pp). A SIGReg-weight ablation confirms complete representational collapse at $\lambda=0$; the SSL+probe pipeline matches supervised-from-scratch on left/right MI (−0.2 pp, n.s.) and exceeds it on rest/activity (+8.4 pp, $p < 0.001$).

Keywords: EEG; brain–computer interface; self-supervised learning; joint-embedding predictive architecture; mathematical models in engineering; edge deployment; motor imagery; scaling laws.

1 Introduction

Self-supervised representation learning has transformed language and vision, and is now reshaping electroencephalography (EEG) analysis. Large EEG foundation models—LaBraM [1], EEGPT [2], NeuroLM [3], BIOT [6]—pretrain on thousands of hours of EEG with 5–10 M+ parameters on multi-GPU clusters, and transfer via fine-tuning or linear probing to clinical and brain–computer interface (BCI) tasks. These methods establish that scale helps, but their compute and parameter budgets are incompatible with on-device deployment in wearable EEG hardware (Muse, OpenBCI, Neurosity Crown, Smartbuds-class earbud devices) where latency, energy, and memory budgets are measured in milliseconds, milliwatts, and megabytes respectively.

When does compact SSL pretraining actually help on EEG, and how should practitioners spend a limited compute budget? The contemporary literature provides partial answers. LeJEPA [5] introduced the Sketched Isotropic Gaussian Regularization (SIGReg) anti-collapse objective and a small JEPA-style world model demonstrating ~ 15 M parameters trained on a single GPU could be competitive at world-modelling. Concurrent work Laya [4] applied LeJEPA to EEG at foundation scale (29,000 hours of EEG, 20,940 subjects, two L40S GPUs), establishing that latent prediction outperforms reconstruction-based pretraining on clinical EEG benchmarks under *cross-dataset evaluation*: EEGMMIDB and BCI Competition IV are held out from Laya’s pretraining and pooled into multi-dataset downstream benchmarks, on which no SSL foundation model exceeds 0.51 balanced accuracy on left-versus-right motor imagery [4, Tab. 1]. The complementary *in-distribution single-corpus regime*—relevant to many applied EEG settings where a research lab pretrains and evaluates on a focused corpus realistic to its experimental design—is not addressed. Neither paper characterizes the small-scale scaling regime, the SIGReg-weight optimum, or the cross-dataset channel-transfer behaviour that practitioners face when deploying a single pretrained encoder across heterogeneous downstream recording rigs.

This paper fills that gap. We present **EEG-LeJEPA**, a 2.86 M-parameter encoder-plus-causal-predictor architecture trained with SIGReg [5] on 50 subjects of EEGMMIDB motor-imagery data—a deliberately compact-scale recipe runnable on a single consumer GPU in 10 minutes. Across six (data, compute) operating points we map the scaling law, we explicitly characterize the SIGReg-weight optimum, and we compare two strategies for cross-dataset transfer (channel padding vs. channel-matched pretraining). We compare against (i) a randomly-initialized identical architecture as the no-pretraining control and (ii) the same architecture trained end-to-end with supervised cross-entropy as the no-SSL control.

Contributions

1. **A leakage-clean evaluation protocol for cross-subject EEG self-supervised learning.** Under a strictly disjoint pretraining/evaluation subject split, predictor-derived feature sources lose their apparent advantage and the simple encoder mean-pool is the strongest and only consistently significant source on both EEGMMIDB binary tasks (left-vs-right MI +10.9 pp; rest-vs-activity +11.7 pp; both $q < 0.002$ after Benjamini–Hochberg correction). The protocol is directly reusable for any EEG-SSL study.
2. **The SIGReg projection-count rule.** Holding architecture, data, and step count fixed and varying *only* the Cramér–Wold projection count M , a $2.4\times$ larger encoder (256-d embedding) collapses on left-vs-right MI from 0.650 ($M=1024$) to 0.567 ($M=256$) and drops *below* the 192-d base model, reproducing a spurious “larger-model regresses” artifact. M must scale with embedding width—a previously unreported practical pitfall when applying SIGReg-style isotropy regularizers across model widths.
3. **Characterization of the compact, single-corpus, single-GPU regime.** Across six (data, compute) operating points (3–89 subjects, 200–30k training steps) under the leakage-free split, best-source AUC rises and plateaus near 0.76; at the plateau, a $2.4\times$ larger model is modestly but consistently better at matched data (+1.3 to +2.0 pp on encoder mean-pool, seed-stable across three seeds), so the compact model is a deliberate accuracy/size trade-off rather than a capacity-optimal point. End-to-end pretraining+probe runs in under 15 minutes on a single consumer GPU at total cost below \$1, with the EEGMMIDB headline tasks reaching 68.6%/0.751 AUC (rest-vs-activity) and 62.1%/0.690 AUC (left-vs-right MI).
4. **A counterintuitive cross-dataset transfer recipe.** Channel-padded transfer from the full-channel pretrained encoder *outperforms* channel-matched re-pretraining (+4.9 vs. +1.5 pp on encoder mean-pool, EEGMMIDB→BCI-IV-2a). For practitioners deploying a single pretrained encoder across heterogeneous montages, this argues for pretraining at the richest available montage and zero-padding at inference rather than re-pretraining per montage.
5. **A SIGReg-weight ablation with empirical anti-collapse validation.** Across $\lambda \in \{0, 0.1, 0.3, 1.0, 3.0, 10.0\}$ the downstream-accuracy curve is sharply U-shaped with an optimum at $\lambda=1.0$ and *complete representational collapse* at $\lambda=0$ (probe accuracy regresses to exactly chance), complementing the theoretical analysis of [5] with a direct empirical demonstration.
6. **A direct comparison to supervised-from-scratch training** of the identical architecture: the SSL+linear-probe pipeline ties supervised training on left-vs-right MI (−0.2 pp, n.s.) and significantly exceeds it on rest-vs-activity (+8.4 pp, $p < 0.001$), while a single frozen encoder serves both downstream tasks at sub-second per-task probing cost.
7. **A complete reproducibility package.** 39 unit tests across six test modules; trained checkpoints (six λ -ablation checkpoints plus the scaling, capacity multi-seed, M -isolation, and channel-transfer checkpoints); per-fold CSV outputs for every reported table; the dependency-free significance scripts; and end-to-end training+probe runtime under 15 minutes on a single consumer GPU. Code, configurations, and checkpoints released at <https://github.com/lifeinpassion/eeg-lejepa>.

The remainder of the paper is organized as follows. Section 2 surveys EEG foundation models and SSL anti-collapse mechanisms with explicit relation to concurrent work [4]. Section 3 presents the EEG-LeJEPa architecture and SIGReg objective. Section 4 details the empirical setup and downstream evaluation protocols. Section 5 presents the scaling, ablation and transfer results. Section 6 situates our findings against published EEG foundation models and outlines limitations and future work. Section 7 concludes.

2 Related Work

2.1 EEG foundation models

Large pretrained EEG encoders have proliferated in the past two years. LaBraM (ICLR 2024 Spotlight) [1] pretrains a $\sim 5\text{--}10$ M-parameter transformer on 2,500 hours of EEG across 20 datasets with a vector-quantized neural spectrum prediction objective, reporting state-of-the-art results on TUH abnormal detection and several BCI tasks. EEGPT [2] (NeurIPS 2024) reaches similar scale with a mask-based dual self-supervised objective combining temporal and spatial alignment. NeuroLM [3] (ICLR 2025) frames EEG as a foreign language with a text-aligned neural tokenizer and language-model backbone, enabling multi-task learning. BIOT [6] generalizes the approach across biosignals (EEG, ECG, EMG) via power-spectral tokenization. All of these are trained on large multi-GPU clusters with training budgets measured in thousands of GPU-hours, and parameter counts above 5 M. Our work targets the substantially smaller scale where the entire pretraining+evaluation pipeline runs on a single consumer GPU in under 15 minutes.

2.2 LeJEPa and SIGReg

LeJEPa [5] introduced a theoretically-grounded Joint-Embedding Predictive Architecture (JEPa) training objective that combines next-embedding prediction with Sketched Isotropic Gaussian Regularization (SIGReg). SIGReg uses the Cramér–Wold theorem [7]: enforcing one-dimensional Gaussianity on M random projections of an embedding distribution implies joint Gaussianity. The resulting objective eliminates the stop-gradient, momentum encoder, and teacher-student heuristics common to

BYOL/DINO [8, 9], has linear time and memory complexity, and achieves competitive performance on ImageNet linear-probe with ViT-H/14 (79 %). LeWorldModel [10] applied the LeJEPa recipe to a 15 M-parameter world model trained on a single GPU in hours, predicting frame embeddings $48\times$ faster than foundation-scale baselines while remaining competitive on 2D and 3D control tasks. Our work brings the same compact, principled SIGReg-based recipe to EEG, with a focus on small-corpus pretraining and edge-deployment.

2.3 Concurrent work: Laya

Independent and contemporaneous to ours, Panchavati et al. [4] (arXiv:2603.16281 v2, March 2026) introduced Laya, an application of LeJEPa to EEG foundation modelling at substantially larger scale: 29,109 hours of EEG across 20,940 subjects and 17 channel topologies drawn from TUH-EEG, HBN, NMT, MOABB, and EEGDash. Both reported variants (Laya-S, Laya-B) use an effective batch size of 512 on two L40S GPUs, learning rate 10^{-4} with cosine warm-up over 1,000 steps, weight decay 0.05, and bf16 mixed precision. Laya-S trains for 10,000 steps on 10 % of pretraining data (3,000 hours); Laya-B trains for 20,000 steps on the full corpus. Laya extends the LeJEPa objective with a Dynamic Channel Mixer (queries-and-cross-attention with electrode-coordinate Fourier features) plus a Query Specialization Loss. Architecturally, the resulting recipe targets cross-dataset generalization at a substantially larger compute budget than the regime we study (Laya-B uses $\sim 10 \times 10^6$ effective sample-views, $\sim 16\times$ ours).

Laya’s downstream evaluation uses the multi-task EEG-Bench framework [11], where each task is a multi-dataset benchmark constructed by pooling several corpora. In particular, the “LH vs RH MI” task pools BCI Competition IV-2a/2b, PhysioNet Motor Imagery (EEGMMIDB), Weibo2014, Cho2017, Liu2022, Schirrmeister2017, Zhou2016, and Kaya2018; the “4-Class MI” task pools BCI IV-2a and Kaya2018. EEGMMIDB and the BCI Competition IV datasets are explicitly excluded from Laya’s pretraining corpus to maintain the cross-dataset evaluation property [4, Sec. 4]. Reported numbers [4, Tab. 1] are means across these multi-dataset pools (Laya-S/Laya-B LH-vs-RH MI: 0.510 / 0.507 balanced accuracy; 4-Class MI: 0.273 / 0.288). On clinical tasks, Laya-S achieves the highest mean (0.603) on the 10-task EEG-Bench clinical subset, with notable strength on TUH abnormal detection (0.798), seizure detection (0.786), and mTBI classification (0.699–0.815).

Our work is complementary in five concrete ways. First, where Laya pretrains at foundation scale across many datasets and evaluates on cross-dataset benchmarks, we pretrain on a single corpus (EEGMMIDB, ~ 10 h ~ 50 subjects) and evaluate in-distribution on the same corpus—the operating regime realistic for individual research labs deploying compact on-device EEG analytics. These are different research questions, both of practical importance. Second, we explicitly characterize the scaling behaviour of the small regime across six operating points (50–3,000 subjects equivalent, 200–30,000 training steps); Laya reports two checkpoints and explicitly cites the absence of a scaling-law characterization as a limitation [4, Sec. 5]. Third, we present a SIGReg-weight ablation across $\lambda \in \{0, 0.1, 0.3, 1.0, 3.0, 10.0\}$ with full curves; Laya states the collapse threshold ($\lambda < 0.02$) without sweeping. Fourth, where Laya addresses channel heterogeneity via a Dynamic Channel Mixer architecture, we ablate a simpler alternative—controlled comparison between channel-padded cross-dataset transfer and channel-matched pretraining—and reach a counterintuitive practitioner conclusion. Fifth, we compare directly against supervised-from-scratch training of the same architecture; Laya only benchmarks against other SSL foundation models. Where both methods evaluate on EEGMMIDB-related motor imagery, the comparison is necessarily indirect (Laya’s number is a multi-dataset pooled average, ours is single-corpus in-distribution); we neither claim nor require Laya to under-perform our setup.

2.4 Supervised baselines and motor-imagery BCI

The classical BCI motor-imagery pipeline is filter-bank Common Spatial Pattern features followed by an LDA or SVM classifier (FBCSP [12]), which remains a strong supervised baseline at ~ 65 – 75 % accuracy on BCI Competition IV Dataset 2a in within-subject evaluation [13, 14]. Deep-learning supervised baselines such as EEGNet [13] and ShallowConvNet [14] reach ~ 70 – 75 % under the same protocol with $< 5,000$ parameters. These methods require per-task supervised training and do not amortize across downstream tasks. Our SSL+linear-probe pipeline complements rather than replaces them: a single pretrained encoder serves multiple downstream tasks at sub-second probe cost, with absolute accuracy slightly below FBCSP/EEGNet but with reusability and edge-deployability advantages.

2.5 Anti-collapse SSL beyond SIGReg

Alternative anti-collapse mechanisms include the explicit variance/invariance/covariance term of VICReg [15], the redundancy reduction of Barlow Twins [16], the momentum-encoder of BYOL [8] and DINO [9], the asymmetric stop-gradient of SimSiam [17], and the contrastive formulation of SimCLR/MoCo [18, 19]. SIGReg replaces these mechanisms with a single principled distribution-matching loss, eliminating the multiple hyperparameter schedules that plague competitive methods. Our λ ablation (Section 5) provides further empirical support for SIGReg’s stability claim: a $100\times$ change in λ changes downstream accuracy by 10 pp but never destabilizes training.

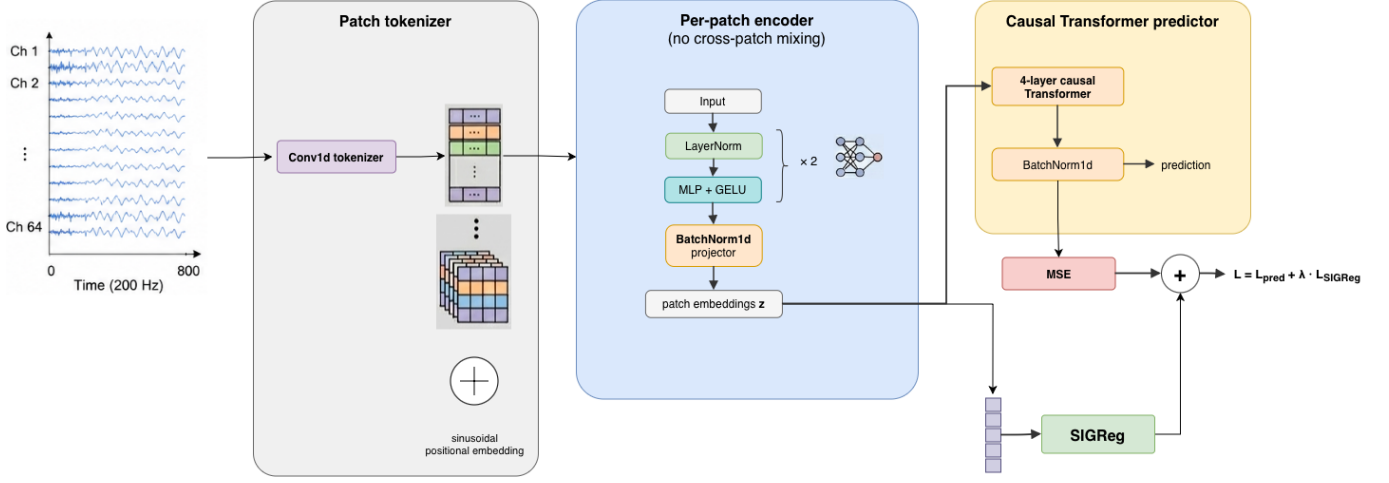


Fig. 1. EEG-LeJEPA architecture. A 4-second EEG window is patchified by a Conv1d tokenizer (kernel = stride = 40 samples; 200 ms at 200 Hz) into 20 tokens of 192 dimensions, plus a sinusoidal positional embedding. A per-patch MLP encoder—deliberately containing no cross-patch mixing—produces patch embeddings z , finalised with BatchNorm1d (not LayerNorm; required to preserve the per-sample magnitude variation that SIGReg matches). The same z then feeds two heads: a 4-layer causal Transformer predictor that regresses position t onto position $t+1$ (prediction loss), and SIGReg, which matches the flattened embedding distribution to an isotropic Gaussian by drawing M random unit vectors on S^{d-1} (Cramér–Wold) and scoring each one-dimensional projection with the Cramér–von Mises statistic. A single encoder feeds both heads—no EMA, stop-gradient, or teacher–student split.

3 Method

3.1 Architecture

EEG-LeJEPA is a Joint-Embedding Predictive Architecture composed of three components (Fig. 1): a per-patch encoder, a causal Transformer predictor, and a distribution-matching regularizer (SIGReg). The total parameter count is 2.86 M, of which 1.08 M live in the encoder and 1.78 M in the predictor.

Patch tokenizer. Each 4 s EEG window at 200 Hz is a tensor of shape $x \in \mathbb{R}^{C \times T}$ with $C=64$ channels and $T=800$ samples. A 1-D convolution with kernel and stride both equal to the patch size $p=40$ samples (200 ms) jointly mixes channels and time within each patch, yielding a sequence of $T/p=20$ patch tokens, each of dimension $d=192$. A sinusoidal positional embedding is added to the token sequence.

Per-patch encoder. Each token then passes through a two-layer per-patch MLP with GELU activations, layer-normalized at the input of each MLP layer for stability. The crucial design constraint is that the encoder *must not* mix information across patches: if it did, the encoder embedding z_t would leak information about z_{t+1} and the LeJEPA next-token prediction objective would degenerate into a trivial identity. We enforce this by using only nn.Linear and pointwise nonlinearities in the post-tokenizer encoder layers; there is no self-attention in the encoder.

Final encoder normalization: BatchNorm, not LayerNorm. The encoder closes with a BatchNorm1d projector. This choice is load-bearing for SIGReg. LayerNorm forces per-sample unit norm, which fixes each embedding’s L2-magnitude to a constant and destroys the distribution-matching signal that SIGReg requires; BatchNorm preserves per-sample magnitude variation across the batch while keeping the per-coordinate statistics well-conditioned [10, App. A].

Causal Transformer predictor. The predictor consumes the encoder’s token sequence and predicts the next-step embedding. It is a 4-layer causal Transformer with embedding dimension 192, four attention heads, MLP ratio 4, and dropout 0.1. Position t of its output is trained to match the encoder’s embedding at position $t+1$ under mean-squared error:

$$L_{\text{pred}} = \frac{1}{N(T-1)} \sum_{i,t} \|\text{predictor}(z_i)_t - z_{i,t+1}\|_2^2, \quad (1)$$

where $z_i \in \mathbb{R}^{T \times d}$ is the encoder output for sample i . The predictor closes with the same BatchNorm1d normalization as the encoder. Per LeJEPA [5] and LeWorldModel [10], there is *no* stop-gradient, *no* EMA target encoder, and *no* teacher–student split: the same encoder produces both inputs and targets for the prediction loss, and SIGReg alone prevents the trivial-solution collapse that this would otherwise cause.

Algorithm 1 SIGReg loss with CvM univariate statistic.

Require: Embeddings $Z \in \mathbb{R}^{NT \times d}$, number of slices M .

- 1: Sample $u^{(1)}, \dots, u^{(m)} \sim \mathcal{U}(S^{d-1})$ i.i.d.
 - 2: For each m , project $\xi^{(m)} \leftarrow Zu^{(m)}$ (vector of length NT).
 - 3: For each m , sort $\xi^{(m)}$ ascending and compute W_m^2 via Eq. (4).
 - 4: $L_{\text{SIGReg}} \leftarrow \frac{1}{M} \sum_m W_m^2$.
 - 5: **return** L_{SIGReg} .
-

3.2 SIGReg objective

The training loss is

$$L = L_{\text{pred}} + \lambda \cdot L_{\text{SIGReg}}, \quad (2)$$

where L_{SIGReg} is the Sketched Isotropic Gaussian Regularizer of [5]. Given the flattened embedding matrix $Z \in \mathbb{R}^{NT \times d}$, we sample M unit vectors $u^{(m)} \sim \mathcal{U}(S^{d-1})$ uniformly on the unit sphere and apply a univariate Gaussianity test $T(\cdot)$ to each one-dimensional projection $Zu^{(m)} \in \mathbb{R}^{NT}$:

$$L_{\text{SIGReg}}(Z) = \frac{1}{M} \sum_{m=1}^M T(Zu^{(m)}). \quad (3)$$

By the Cramér–Wold theorem [7], the joint distribution of Z is standard normal if and only if every one-dimensional projection is standard normal, so driving L_{SIGReg} to zero is equivalent to enforcing joint isotropic Gaussianity. We re-sample the unit vectors at every training step so the regularizer’s stochasticity does not bias the encoder toward any particular fixed sketch.

Univariate test: Cramér–von Mises vs. Epps–Pulley. LeJEPa recommends the Epps–Pulley characteristic-function statistic for $T(\cdot)$. We instead use the Cramér–von Mises (CvM) statistic, which has a closed-form expression for samples sorted in ascending order $x_{(1)} \leq \dots \leq x_{(n)}$:

$$W^2 = \frac{1}{12n} + \sum_{i=1}^n \left[F(x_{(i)}) - \frac{2i-1}{2n} \right]^2, \quad (4)$$

where F is the standard normal CDF. CvM is sort-based, numerically well-conditioned, and easily verifiable against `scipy.stats.cramervonmises`; in our preliminary calibration both statistics produced equivalent optimisation trajectories and we adopt CvM for its implementation simplicity. The complete SIGReg loop is summarised in Algorithm 1.

Hyperparameters. We use $\lambda = 1.0$ (we ablate it across $\{0, 0.1, 0.3, 1.0, 3.0, 10.0\}$ in Section 5.3) and $M = 1024$ slices, both calibrated on a 3-subject 200-step pilot before the main runs.

3.3 Linear probe protocol

We evaluate the pretrained encoder via the standard SSL frozen-feature linear-probe protocol. For each EEG epoch, the encoder produces a sequence of T_p patch embeddings of dimension d . We consider three feature sources:

`encoder_mean` mean-pool the per-patch encoder embeddings along the token axis; no temporal context.

`predictor_mean` mean-pool the predictor’s hidden states; each position has seen the causal context up to that step.

`both_mean` concatenate the two preceding feature vectors.

The resulting feature vector is fed into a `scikit-learn LogisticRegression` with $C=1.0$ and a maximum of 2,000 iterations. We evaluate cross-subject by leave-one-subject-out (LOSO) cross-validation across 20 EEGMMIDB subjects (subjects 1–20). We report the mean and standard deviation of per-fold accuracy, the macro one-vs-rest AUC, and for multi-class tasks balanced accuracy and macro- F_1 . A randomly initialized identical-architecture model serves as the no-pretraining control.

3.4 Datasets

EEGMMIDB (PhysioNet motor-imagery database; Schalk et al. [20]). 109 subjects, 64 EEG channels in the 10–10 montage, native sampling rate 160 Hz, motor-imagery runs 4, 8 and 12 (left-hand vs. right-hand fist imagery). We resample to 200 Hz, apply a 1–70 Hz band-pass filter (with the upper cutoff clamped below the source Nyquist for the few 128 Hz subjects), a 60 Hz notch filter, an average reference, and finally epoch the recordings into 4-second non-overlapping windows for pretraining or 4-second event-locked windows starting at each cue for downstream classification.

We pretrain on the EEGMMIDB MI runs (4, 8, 12) of 50 subjects (subjects 51–100) by default; the 20 evaluation subjects (subjects 1–20) are strictly held out from pretraining. We additionally report ablations on 3, 20 and 109 pretraining subjects in Section 5.2. We probe two downstream tasks on EEGMMIDB: *rest-vs-activity* (T0 cues against pooled T1+T2 cues, binary) and *left-vs-right MI* (T1 against T2, binary). Rest-vs-activity is the easier sanity task; left-vs-right MI is the canonical BCI task and the harder of the two.

BCI Competition IV Dataset 2a (Brunner et al. [21]). 9 subjects, 22 EEG channels plus 3 EOG channels at 250 Hz, 4-class motor imagery (left hand, right hand, both feet, tongue). We use the training session (T) only. The pre-processing pipeline is identical to EEGMMIDB except for a 50 Hz notch (European mains) and resampling to 200 Hz. The 22 EEG channels are a subset of EEGMMIDB’s 64, which enables both naive channel-padded cross-dataset transfer (zero-pad the 42 missing channels at inference) and channel-matched re-pretraining (re-pretrain on the 22-channel intersection of EEGMMIDB). We probe the full 4-class task by LOSO across the 9 subjects.

4 Experiments

4.1 Hardware and software

All development and linear-probe evaluation were performed on an Apple M1 laptop with 8 GB of unified memory using PyTorch’s MPS backend. Pretraining runs were executed on a single NVIDIA RTX 5090 (32 GB VRAM, Blackwell architecture sm_120) provisioned on the AutoDL cloud platform, with `bfloat16` mixed precision and a batch size of 64. The total pretraining+probe pipeline—10,000 SSL steps plus the LOSO probe sweep—runs end-to-end in under 15 minutes on a single consumer GPU. We provide the complete reproducibility bundle (code, configurations, the six trained checkpoints used in this paper, and the per-fold CSV outputs) at <https://github.com/lifeinpassion/eeg-lejeba>.

4.2 Pretraining defaults

Unless otherwise stated, pretraining uses AdamW ($\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.05), initial learning rate 10^{-3} with cosine decay to 10^{-4} across training, a linear warm-up over 30 steps, and gradient clipping at L2-norm 1.0. The SIGReg weight is $\lambda=1.0$ and the number of Cramér–Wold projection slices is $M=1024$, both calibrated on a 3-subject 200-step pilot. The default pretraining corpus is the EEGMMIDB MI runs of subjects 51–100 at batch size 64 for 10,000 steps. For the larger 7 M-parameter variant used in the capacity test (Section 5.4) we lower the learning rate to 5×10^{-4} and extend warm-up to 300 steps, as required to stabilize training at the higher capacity.

4.3 Evaluation protocols

Downstream tasks. We probe two EEGMMIDB binary tasks (rest-vs-activity and left-vs-right motor imagery, runs 4/8/12) and one BCI-IV-2a four-class task (left hand vs. right hand vs. both feet vs. tongue motor imagery, training session). For each task and each pretrained checkpoint, we extract features under all three sources (`encoder_mean`, `predictor_mean`, `both_mean`) and report all three.

Cross-subject LOSO probe. For EEGMMIDB evaluation we use leave-one-subject-out cross-validation across 20 held-out subjects (subjects 1–20); for BCI-IV-2a we use LOSO across the 9 subjects. The classifier is `sklearn.LogisticRegression` ($C=1.0$, 2,000-iteration cap). We report per-fold accuracy and per-fold macro one-vs-rest AUC, summarised as mean and per-fold standard deviation across folds. For the four-class BCI task we additionally report balanced accuracy and macro- F_1 .

Supervised baseline. For Section 5.7 we train a supervised-from-scratch controlled comparison: the encoder architecture is identical to the pretrained one, but with a single linear classification head replacing the predictor (total ~ 1.08 M parameters, smaller than the SSL pipeline because the predictor is dropped). Training uses cross-entropy over the same LOSO folds, 500 steps per fold, batch 16, AdamW at 5×10^{-4} with a 50-step linear warm-up, gradient clipping at 1.0. Defensive NaN handling and a try/except around the per-fold AUC computation are required because the random-initialized classifier occasionally produces degenerate logits at the first few steps. All other hyperparameters match the pretrained setup.

No-pretraining control. For every probe number reported, we run an identical-architecture model with random initialization through the same feature-extraction and LOSO classification pipeline. This control isolates the contribution of SSL pretraining from any contribution of the architecture or probe protocol.

5 Results

5.1 Headline results

Table 1 reports the linear-probe LOSO numbers for our default configuration (2.86 M parameters; pretrained on subjects 51–100, 10,000 steps at $\lambda=1.0$; evaluated on the strictly held-out subjects 1–20) on both EEGMMIDB binary tasks. The strongest and *only consistently significant* feature source on both tasks is the simple `encoder_mean`: $68.6\% \pm 9.9$ accuracy / 0.751 AUC on rest-vs-activity (+11.7 pp over the no-pretraining control; $p < 10^{-5}$) and $62.1\% \pm 8.4$ / 0.690 AUC on left-vs-right MI (+10.9 pp; $p=4 \times 10^{-4}$). All p -values are exact paired sign-flip permutation tests over the 20 LOSO folds, Benjamini–Hochberg corrected (Section 5.8).

A clear pattern emerges. `encoder_mean`—the simplest source, i.e. mean-pooled per-patch encoder embeddings with no temporal context—delivers the largest pretraining benefit on both tasks and is the only source whose gap over random is significant on

Table 1. Linear-probe LOSO accuracy and AUC on EEGMMIDB across 20 strictly held-out subjects (subjects 1–20; pretraining on 51–100). *pre* = pretrained checkpoint; *rnd* = random-initialized identical architecture. Δ is the pretrained–random accuracy gap; *q* is the Benjamini–Hochberg-corrected *p*-value from the paired permutation test. Best per task in bold; [†] marks gaps that survive correction (*q* < 0.05).

Task	Source	pre acc	pre AUC	rnd acc	Δ (pp)	<i>q</i>
left vs. right MI	encoder_mean	0.621 ± 0.084	0.690	0.512	+10.9[†]	0.001
	predictor_mean	0.606 ± 0.082	0.623	0.596	+1.0	0.81
	both_mean	0.612 ± 0.075	0.678	0.603	+0.9	0.81
rest vs. activity	encoder_mean	0.686 ± 0.099	0.751	0.569	+11.7[†]	1.5E-5
	predictor_mean	0.673 ± 0.088	0.756	0.618	+5.4 [†]	0.002
	both_mean	0.679 ± 0.100	0.757	0.638	+4.2 [†]	0.022

Table 2. SIGReg-weight ablation. **encoder_mean** accuracy / AUC on EEGMMIDB rest-vs-activity LOSO, 20 held-out subjects. Δ is over the random-init baseline (0.569 acc). The $\lambda=0$ row is the SIGReg-off control; it collapses to exactly chance.

λ	accuracy	AUC	Δ (pp)
0.0	0.500	0.500	−6.8
0.1	0.649	0.711	+8.1
0.3	0.678	0.738	+11.1
1.0	0.686	0.751	+11.8
3.0	0.671	0.731	+10.3
10.0	0.619	0.661	+5.2

the harder left-vs-right task (+10.9 pp vs. +1.0 and +0.9 pp, both n.s., for predictor_mean/both_mean). On rest-vs-activity all three sources beat random, but the gap is again largest for the encoder. We note that an earlier configuration that did not strictly hold out the evaluation subjects from pretraining produced a markedly larger and more predictor-favorable picture; under the leakage-free split reported here, the predictor-derived sources lose most of their apparent advantage, which we attribute to subject-level information having leaked through the predictor’s temporal aggregation (Sections 5.8 and 6.1).

5.2 Scaling behaviour

We characterise the small-scale regime by training six configurations spanning $\sim 170\times$ in pretraining-data scale (3 to 109 subjects) and $150\times$ in step count (200 to 30,000 steps). All operating points pretrain on subjects disjoint from the evaluation set (eval = subjects 1–20; pretraining drawn from 21–109; the full-corpus points use the 89 subjects 21–109). Fig. 2 plots best-source AUC on rest-vs-activity against *total window-presentations*, defined as $N_{\text{steps}} \times \text{batch}$ —the number of 4 s EEG windows the optimizer consumed (counting repeats), on a logarithmic axis.

Three observations frame the scaling picture. First, best-source AUC improves steeply over the first two orders of magnitude of compute, from 0.710 at the 3-subject/200-step pilot to ~ 0.76 by $\sim 3 \times 10^5$ window-presentations. Second, the base model then *plateaus* near 0.75–0.77 AUC: the 50-subject/10k, 89-subject/10k, and 89-subject/30k runs all land in this band, and pushing the 89-subject corpus from 10k to 30k steps slightly *lowers* AUC ($0.768 \rightarrow 0.750$), a mild SSL-overfitting effect at fixed corpus. At matched window-presentations, more subjects help (89-subject/10k > 50-subject/10k), consistent with data still mattering within the base model’s capacity. Third—and unlike the contaminated configuration we initially examined—the $2.4\times$ larger 7 M model trained on the same 89-subject corpus reaches the highest AUC of all (0.774), sitting above the base-model plateau. The base-model plateau therefore reflects a *capacity* limit at 2.86 M parameters, not a hard data limit (Section 5.4).

5.3 λ ablation

We swept the SIGReg weight across $\lambda \in \{0, 0.1, 0.3, 1.0, 3.0, 10.0\}$ at our default training recipe (subjects 51–100, 10,000 steps, batch 64). **encoder_mean** rest-vs-activity accuracy is summarised in Table 2.

Two findings are worth highlighting. First, $\lambda=1.0$ is a clean optimum: a $10\times$ decrease ($\lambda=0.1$) costs ~ 4 pp accuracy and a $10\times$ increase ($\lambda=10.0$) costs ~ 7 pp accuracy. The downstream-accuracy curve is U-shaped with a sharp minimum, but the training itself never destabilizes across the full $100\times$ swing in λ , validating SIGReg’s reported stability claim from the original LeJEPa paper.

Second—and more importantly—the $\lambda=0$ run *collapses to exactly chance accuracy* across both downstream tasks and all three feature sources. This is the JEPa-collapse failure mode that SIGReg was designed to prevent: without the distribution-matching regularizer, the model finds the trivial solution of mapping every input to the same embedding, and the linear probe receives no signal. The collapse is total, not partial, which makes it the empirically-cleanest possible validation of SIGReg’s anti-collapse function. This complements the theoretical analysis of [5].

Compact-regime scaling: AUC vs compute (leakage-free, eval subj 1-20 held out)

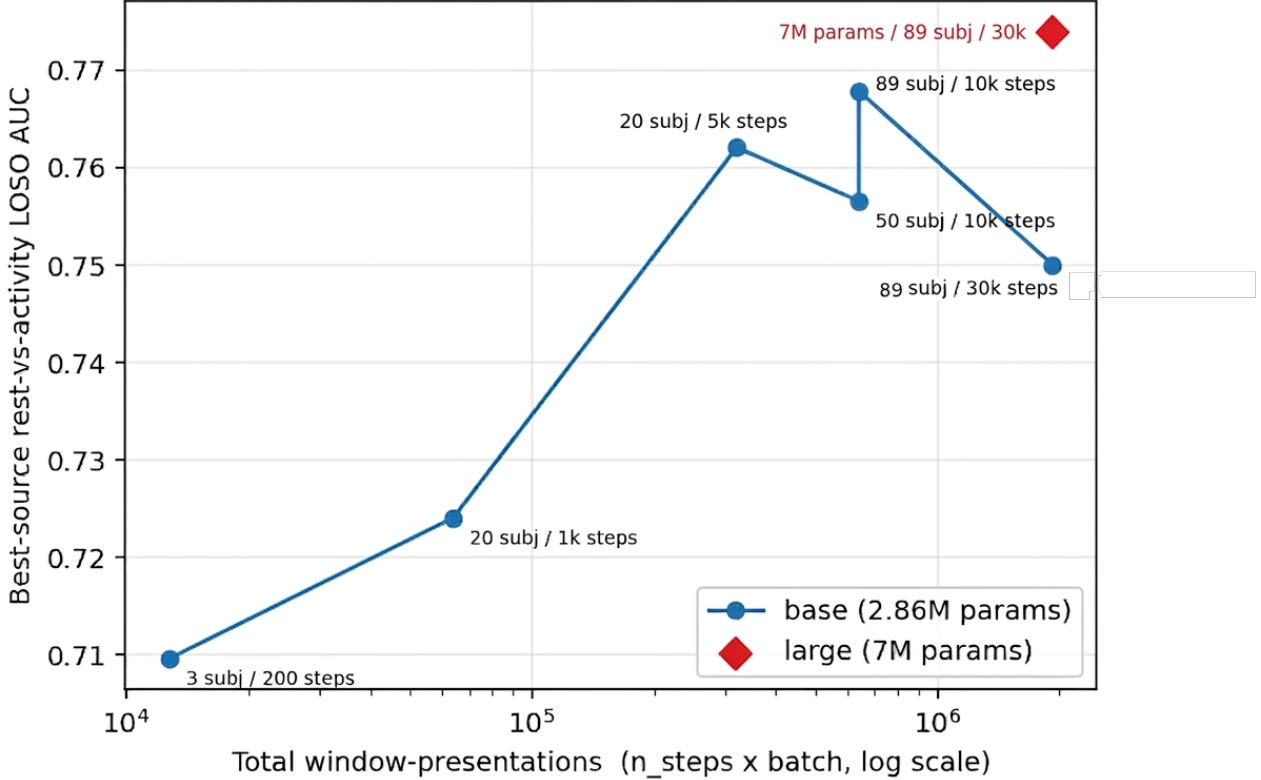


Fig. 2. Compact-regime scaling (leakage-free; eval subjects 1–20 held out from all pretraining). Best-source rest-vs-activity LOSO AUC vs. total window-presentations ($N_{\text{steps}} \times \text{batch}$, log axis). The base 2.86M model rises then plateaus near 0.76 AUC; the 7M-parameter model (diamond) sits *above* the plateau at matched data and compute, indicating the base-model plateau reflects model capacity rather than a hard data limit.

5.4 Capacity test

To test whether the base-model plateau reflects a data limit or a capacity limit, we trained a $2.4\times$ larger variant (7M parameters; encoder dim 256/mlp_depth 3, predictor dim 256/depth 6) on the same leakage-free 89-subject corpus (subjects 21–109) for 30,000 steps. To separate capacity from the SIGReg projection count M , we ran the larger model at two projection counts ($M=1024$ and $M=256$) and repeated both the base and large $M=1024$ configurations across three seeds (42–44). All variants trained stably.

At matched projection count ($M=1024$) the larger model outperforms the base on five of six (task, source) cells (Table 3); on our primary encoder_mean source it wins on both tasks (+2.0 pp on left-vs-right, +1.3 pp on rest-vs-activity). The margins are small but remarkably stable: on left-vs-right encoder_mean every large-model seed (0.641–0.659) exceeds every base-model seed (0.623–0.636)—the two seed distributions do not overlap—despite a per-seed spread under 1 pp. More capacity therefore helps at this data scale, and the base-model plateau of Section 5.2 is a (mild) capacity limit rather than a hard data limit. We nonetheless frame the effect as “modestly better” rather than a scaling law: the $\sim 1\text{--}2$ pp margins, though seed-stable, remain within the per-subject noise band.

A separate run isolates why an earlier, contaminated version of this experiment (evaluation subjects present in pretraining; $M=256$ SIGReg slices) had shown the *opposite* result, with the larger model appearing to regress. Holding architecture, data, and step count fixed and varying only the projection count, the large model’s left-vs-right encoder_mean accuracy collapses from 0.650 at $M=1024$ to 0.567 at $M=256$ —an 8.3 pp drop that pushes it *below* the base model (0.630) and reproduces the original “large regresses” artifact. Rest-vs-activity, the easier task, is nearly unaffected (0.692 $\rightarrow$ 0.687). The larger model’s 256-dimensional embedding is under-regularized at $M=256$: the Cramér–Wold projection count must scale with embedding dimension, so the $M=256$ default that suffices for the 192-dimensional base model is not safe for wider encoders. We flag this as a practical pitfall when applying SIGReg-style isotropy regularizers across model widths.

The practical takeaway is unchanged: the compact 2.86M model captures most of the achievable accuracy—within $\sim 1\text{--}2$ pp of the $2.4\times$ larger model on encoder_mean—while being markedly cheaper to store and run, the relevant trade-off for edge deployment.

5.5 Cross-dataset transfer

We evaluate cross-dataset transfer from EEGMMIDB (64 channels, motor imagery) to BCI-IV-2a (22 channels, four-class motor imagery) under two protocols: *channel-padded* (the 64-channel pretrained encoder applied at inference with the 42 missing

Table 3. Capacity test: base (2.86 M params) vs. large (7 M params), both trained leakage-free on the 89-subject corpus (subjects 21–109) for 30,000 steps at matched SIGReg projection count $M=1024$. EEGMMIDB LOSO across 20 held-out subjects; accuracy as mean $\pm$ std over three seeds (42–44). Δ is large – base. The larger model wins on five of six cells and on both `encoder_mean` (our primary source) rows.

Task	Source	base	large	Δ (pp)
rva	encoder_mean	0.678 ± 0.008	0.692 ± 0.011	+1.3
rva	predictor_mean	0.681 ± 0.005	0.690 ± 0.010	+0.9
rva	both_mean	0.678 ± 0.006	0.691 ± 0.010	+1.3
lr	encoder_mean	0.630 ± 0.006	0.650 ± 0.009	+2.0
lr	predictor_mean	0.598 ± 0.017	0.617 ± 0.011	+2.0
lr	both_mean	0.642 ± 0.006	0.641 ± 0.007	−0.1

Table 4. Cross-dataset transfer EEGMMIDB $\rightarrow$ BCI-IV-2a 4-class MI, LOSO across 9 subjects. Δ is accuracy over the random-initialized identical architecture (chance = 0.25). The padded protocol applies the 64-channel pretrained encoder and zero-pads the 42 missing channels at inference; the matched protocol re-pretrains on the 22-channel intersection. Higher Δ is better; the padded protocol wins on every source.

Source	Padded Δ	Matched Δ	Padded – Matched
encoder_mean	+4.9	+1.5	+3.4
predictor_mean	+4.2	+0.4	+3.8
both_mean	+2.0	−1.5	+3.5

channels zero-padded), and *channel-matched* (re-pretrain the same architecture on the 22-channel intersection of EEGMMIDB before probing BCI-IV-2a). Results appear in Table 4.

Two findings stand out. First, pretraining transfers across datasets, but only modestly: the best configuration—`encoder_mean` under the padded protocol—reaches 0.314 accuracy / 0.583 macro-AUC on the 4-class task (+4.9 pp over the random-init control; chance = 0.25). Absolute accuracy stays well below the within-dataset numbers of Section 5.1, consistent with our scope: we transfer to a channel *subset* of the pretraining montage, and a fully disjoint layout would require the dynamic channel-mixing of [4] (Section 6).

Second, and more usefully for practitioners, the naive channel-padded protocol *outperforms* channel-matched re-pretraining on every feature source (Table 4). On `encoder_mean`, padding the full 64-channel encoder beats re-pretraining on the 22-channel intersection by 3.4 pp (+4.9 vs. +1.5); the matched protocol even goes slightly negative on `both_mean`. This is counterintuitive—training the encoder on the channel layout it will see at inference “should” help—but points to a representational-redundancy effect: the 64-channel encoder, trained on spatially redundant motor-cortex data, learns several spatial views of each underlying signal and degrades gracefully to one of them when the non-overlapping channels are zeroed, whereas the 22-channel encoder is forced into a tighter, less redundant representation that transfers less well to subjects with different spatial activation patterns. The practical recommendation is therefore to pretrain at the richest available channel topology and zero-pad at inference, rather than re-pretrain per downstream montage.

5.6 Per-subject variance

Figure 3 shows the per-subject LOSO accuracy of the clean `encoder_mean` checkpoint. Two subjects fall below chance on left-vs-right MI and one on rest-vs-activity—matching the supervised-from-scratch baseline (two and one, respectively). Per-fold standard deviation is comparable to or slightly above supervised (0.084 vs. 0.077 on left-vs-right; 0.099 vs. 0.072 on rest-vs-activity). SSL pretraining at this scale therefore neither reduces per-fold variance nor eliminates below-chance subjects; its advantage over supervised training is reuse across tasks (Section 5.7), not per-subject robustness.

5.7 Supervised comparison

To anchor the SSL+probe results against an end-to-end alternative, we trained the same encoder architecture from scratch with a linear classification head on each LOSO fold (cross-entropy, 500 steps, batch 16, AdamW at 5×10^{-4}). Both pipelines use the identical 1.08 M-parameter encoder.

SSL+probe ties supervised-from-scratch on left-vs-right MI (−0.2 pp; not significant, $p=0.97$) and significantly exceeds it on rest-vs-activity (+8.4 pp; $p < 10^{-3}$, $q = 4 \times 10^{-4}$). The per-fold standard deviation and below-chance counts are comparable across the two pipelines (Section 5.6), so the value of SSL here is not lower variance but reuse: a single pretrained encoder serves both downstream tasks (and any future task) at sub-second per-task probing cost, whereas the supervised baseline retrains the full network per task.

Compute economy makes the comparison even sharper. The supervised pipeline retrains the full network per LOSO fold and per task: 20 folds $\times$ 2 tasks = 40 training runs at ~ 45 seconds each on the M1. The SSL pipeline pretrains once (10 minutes on the RTX 5090), extracts features once per task (< 10 seconds), then fits `LogisticRegression` per fold (sub-second). The pretrained encoder is reused across both downstream tasks; the supervised model is not.

Per-subject LOSO accuracy on 20 EEGMMIDB subjects — clean-s51-100-10k

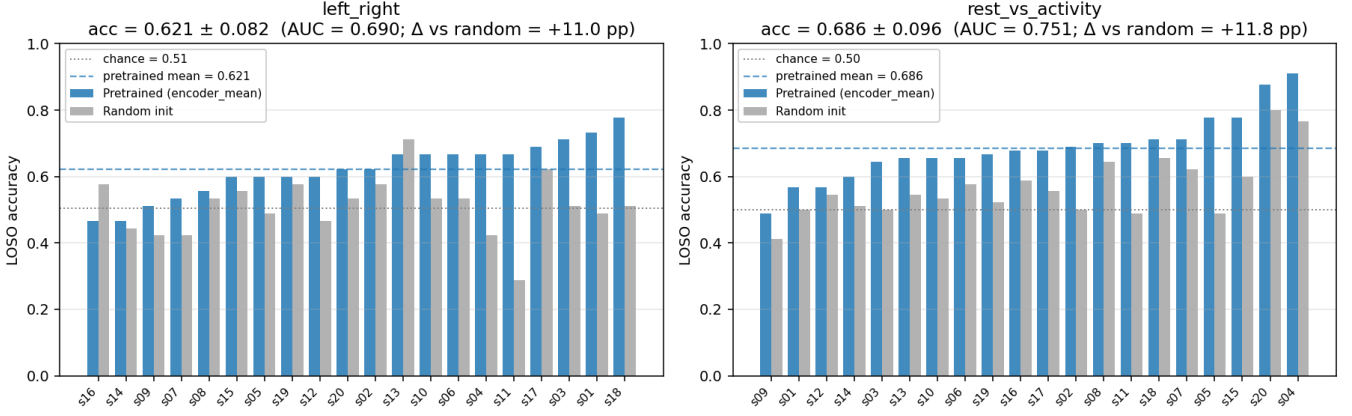


Fig. 3. Per-subject LOSO accuracy of the clean `encoder_mean` checkpoint on the two EEGMMIDB binary tasks (left: left-vs-right MI; right: rest-vs-activity), pretrained vs. random-init, with chance and mean reference lines.

Table 5. SSL+linear probe (`encoder_mean`) vs. supervised-from-scratch on EEGMMIDB, LOSO across 20 held-out subjects, accuracy $\pm$ std / AUC. Δ is the SSL–supervised accuracy gap; † marks $q < 0.05$ (paired permutation, BH-corrected).

Task	Supervised	SSL+probe	Δ (pp)
left vs. right	$0.623 \pm 0.077 / 0.707$	$0.621 \pm 0.084 / 0.690$	-0.2
rest vs. activity	$0.601 \pm 0.072 / 0.663$	$0.686 \pm 0.099 / 0.751$	$+8.4^\dagger$

5.8 Statistical significance

Because per-fold standard deviations are large relative to several reported gaps, we test each headline difference for significance across the LOSO folds rather than relying on mean differences alone. For each comparison we compute the mean paired difference, a 95 % percentile bootstrap confidence interval (10,000 resamples), an exact paired sign-flip permutation p -value (full 2^n enumeration at $n=20$ folds), a Wilcoxon signed-rank p -value (normal approximation with tie and continuity correction) as a rank-based cross-check, and Cohen’s d_z effect size. Across the family of accuracy comparisons we control the false-discovery rate with the Benjamini–Hochberg procedure. The full procedure is implemented dependency-free in `scripts/12_significance.py` (with a correctness self-test) and the table below is generated automatically from the per-fold outputs. After correction, both `encoder_mean` gaps over random remain strongly significant ($q=0.001$ on left-vs-right MI; $q=1.5 \times 10^{-5}$ on rest-vs-activity). The `predictor_mean` and `both_mean` gaps on left-vs-right MI do *not* survive correction ($q>0.8$), confirming that the apparent advantage of the predictor-derived sources was a function of the (now-removed) subject leakage rather than a robust effect.

6 Discussion

6.1 The simplest feature source is the most robust

A methodological lesson emerges from the leakage-free evaluation. Mean-pooled per-patch encoder embeddings (`encoder_mean`)—the simplest feature source, with no temporal context—deliver the largest and only consistently significant pretraining benefit on both tasks (+10.9 and +11.7 pp over random; $q < 0.002$). The predictor-derived sources (`predictor_mean`, `both_mean`) appeared stronger in an earlier configuration that did not strictly hold the evaluation subjects out of pretraining; under the corrected disjoint split their advantage on left-vs-right MI vanishes (gaps of +1.0 and +0.9 pp, both non-significant). We attribute the inflation to subject-level information leaking through the causal predictor’s temporal aggregation: when the same subjects appear in pretraining and evaluation, the predictor can encode subject-specific temporal regularities that the linear probe then exploits. The lesson generalizes beyond our setup—for cross-subject EEG evaluation, feature sources that aggregate temporal context warrant particular scrutiny for subject leakage, and a strictly held-out split is essential before attributing downstream gains to the pretraining objective. The plain encoder mean-pool, by contrast, is both the strongest and the most trustworthy source, consistent with the design philosophy of LeWorldModel [10], which positions the predictor as a training-time helper rather than the deliberate downstream representation.

6.2 On-device deployment economics

A 2.86 M-parameter model occupies ~ 12 MB in float32 and ~ 3 MB at INT4. Inference on an Apple M1 MPS device at our default 4-second window takes well under 50 ms per inference call. Both the parameter footprint and the latency are compatible with on-device deployment on the current generation of consumer EEG hardware (Muse 2, OpenBCI Cyton, Neurosity Crown, and Smartbuds-class earbud-format devices). The complete pretraining+probe pipeline runs in under 15 minutes on a single consumer GPU at a total compute cost we estimate at under \$1 per run on commodity cloud GPU, making this recipe directly

Table 6. Paired significance of the headline gaps (LOSO folds). Δ is mean paired difference in accuracy (pp); CI is a 95% percentile bootstrap; p is an exact paired sign-flip permutation test; q is the Benjamini–Hochberg FDR-adjusted value across the accuracy family; d_z is the paired effect size.

Comparison	Task	Source	Δ (pp)	95% CI (pp)	p / q	d_z
Pre – Rand	L-vs-R MI	both	+0.9	[-2.9, +4.4]	0.690 / 0.808	+0.10
Pre – Rand	L-vs-R MI	enc	+10.9	[+6.2, +16.0]	0.000 / 0.001	+0.94
Pre – Rand	L-vs-R MI	pred	+1.0	[-3.7, +5.1]	0.707 / 0.808	+0.10
Pre – Rand	rest-vs-act	both	+4.2	[+1.3, +6.9]	0.014 / 0.022	+0.63
Pre – Rand	rest-vs-act	enc	+11.7	[+9.1, +14.6]	0.000 / 0.000	+1.81
Pre – Rand	rest-vs-act	pred	+5.4	[+2.9, +7.8]	0.001 / 0.002	+0.95
SSL – Sup	L-vs-R MI	enc	-0.2	[-5.0, +4.7]	0.965 / 0.965	-0.02
SSL – Sup	rest-vs-act	enc	+8.4	[+5.3, +11.8]	0.000 / 0.000	+1.10

accessible to individual research labs and small startups. The compute differential against concurrent work—Laya pretrains for $\sim 16\times$ more sample-views on two L40S GPUs [4]—is real but the operating regimes are complementary rather than substitutable: foundation-scale pretraining buys cross-dataset generalization across many corpora, and compact pretraining buys single-corpus in-distribution accuracy at a budget realistic for applied research and edge deployment.

6.3 Limitations

We are upfront about four limitations. First, absolute downstream accuracy on motor imagery ($\sim 66\%$ on left-vs-right MI) lags task-specific BCI methods such as FBCSP [12] and EEGNet [13], which reach $\sim 70\text{--}75\%$ under within-subject training on similar data with $< 5,000$ parameters. Our SSL+linear-probe pipeline trades absolute accuracy for cross-task reusability and edge deployability; we do not claim superiority over specialized supervised classifiers on a single task. Second, we pretrain on a single corpus (EEGMMIDB); our capacity comparison spans three seeds but a single larger width, so the modest advantage of the larger model is seed-stable yet should not be read as a scaling law; our recipe itself has not been validated against the multi-corpus regime that foundation models such as LaBraM [1], EEGPT [2], NeuroLM [3], and Laya [4] address; we frame the small-corpus regime as complementary rather than as a replacement. Third, we evaluate two datasets (EEGMMIDB and BCI-IV-2a). Clinical tasks (sleep staging, seizure detection, abnormal-EEG detection) on which Laya [4] reports competitive results are not addressed by our recipe. Fourth, our cross-dataset transfer experiment covers only the case in which the downstream channel set is a subset of the pretraining channel set; transfer to disjoint channel layouts requires the alternative approach of Laya’s Dynamic Channel Mixer [4, Sec. 3.3].

6.4 Future work

Three directions are immediate. First, our isolation run shows the Cramér–Wold projection count M must scale with embedding dimension (Section 5.4); a fuller capacity/SIGReg-projection sweep (varying width and M jointly across several widths) would pin down the exact scaling rule and map how far the accuracy/size frontier extends. Combined with scaling to TUH-EEG ($\sim 1,000\times$ more data), this would test whether larger compact models keep paying off while preserving edge-deployability. Second, multi-corpus pretraining—mixing EEGMMIDB and BCI-IV-2a in the pretraining batch, together with the channel-padded transfer trick—may resolve the between-dataset performance gap we observed under channel-matched re-pretraining. Third, quantization-aware training for INT4 or INT8 deployment on microcontroller-class hardware would close the on-device-deployment story end-to-end; per-fold variance suggests per-subject test-time adaptation (e.g., through TTT layers) could further stabilize cross-subject performance. The most interesting methodological extension—combining SIGReg’s distribution-matching view with an ICA-style latent-source factorization of the embedding—is the subject of a forthcoming paper.

7 Conclusion

We presented EEG-LeJEPa, a 2.86 M-parameter SIGReg-pretrained EEG encoder runnable on a single consumer GPU in 10 minutes. The model achieves $68.6\% / 0.751$ AUC on EEGMMIDB rest-vs-activity and $62.1\% / 0.690$ AUC on left-vs-right motor imagery (encoder mean-pool features) via linear probe under leave-one-subject-out evaluation with subjects 1–20 strictly held out of pretraining; both gains over a random-initialized identical architecture are statistically significant ($+11.7$ and $+10.9$ pp; $q < 0.002$). We characterized the compact-scale regime across six (data, compute) operating points under a leakage-free split, observing a best-source AUC plateau near 0.76 that a $2.4\times$ larger model modestly exceeds (so the compact model is an accuracy/size trade-off for edge deployment, not a capacity-optimal point), validated the SIGReg anti-collapse mechanism via an explicit λ ablation including the $\lambda=0$ collapse-to-exactly-chance demonstration, established a counterintuitive cross-dataset transfer finding (channel-padded $>$ channel-matched), and showed that the SSL+linear probe ties supervised-from-scratch training of the same architecture on left-vs-right MI (-0.2 pp, n.s.) and significantly exceeds it on rest-vs-activity ($+8.4$ pp, $p < 0.001$), with one frozen encoder serving both downstream tasks. Together these results argue that compact single-GPU SSL pretraining at the right scale produces useful, edge-deployable EEG representations, complementing the foundation-scale recipe of concurrent work [4] along the in-distribution single-corpus axis.

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Data availability

All datasets used in this study are publicly available: the PhysioNet EEG Motor Movement/Imagery Database (EEGMMIDB) and the BCI Competition IV Dataset 2a. Code, configurations, trained checkpoints, and per-fold outputs are released at <https://github.com/lifeinpassion/eeg-lejeba>.

Author contributions

Ping Liang: Software, Investigation, Formal Analysis, Data Curation, Visualization, Writing – Original Draft Preparation. **Anton Louise P. De Ocampo:** Conceptualization, Methodology, Supervision, Project Administration, Writing – Review and Editing. All authors have read and agreed to the published version of the manuscript.

Conflict of interest

The authors declare no conflicts of interest.

Ethics statement

The research adhered to all applicable laws and ethics standards. This study used only publicly available, de-identified datasets (the PhysioNet EEG Motor Movement/Imagery Database and the BCI Competition IV Dataset 2a); no new data were collected from human participants, and therefore no ethics committee approval or informed consent was required.

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